Manthri Arjun Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.00
Tirumala Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh (BIEAP))	2020	9.69%
Sri Chaitanya English Medium High School	Class X (Board of Secondary Education, Andhra Pradesh (BSEAP))	2018	9.80%

TECHNICAL SKILLS

Analog IC Design: Two-stage OTA, Op-Amp, Bandgap Reference, PTAT/CTAT, Current Mirrors, Bias Circuits, PVT Analysis

EDA Tools: Cadence Virtuoso, Spectre, LTspice, MATLAB

HDL/Programming: Verilog, C

PROJECTS

M.Tech Thesis: High-Precision Bandgap Reference with Curvature Compensation

Thesis Advisor: Dr. Gajendranath Chaudhury | IIT Hyderabad | Ongoing

- Designing a precision CMOS BGR with curvature compensation to minimize temperature-dependent variation in reference voltage.
- Working on PTAT/CTAT generation, temperature compensation, and trimming techniques for improved reference accuracy.
- Evaluating temperature coefficient, PVT variations, and VREF stability using Cadence Virtuoso.

Low-Dropout (LDO) Voltage Regulator Design

Cadence Virtuoso

- Designed a PMOS-based LDO with a two-stage OTA error amplifier and PMOS pass transistor.
- Achieved 66 dB loop gain, 68.5° phase margin, and 0.1327 mV/mA load regulation.
- Performed AC, transient, noise, and PVT-corner simulations for verification.

Bandgap Reference

Analog IC Design Lab | Prof. Ashudeb Dutta | Mar 2026

- Designed an OTA-based Bandgap Reference (BGR) using GPDK 45 nm CMOS for VDD = 1.8 V.
- Achieved VREF = 1.17 V using PTAT/CTAT compensation.
- Designed and optimized OTA, biasing, and current-mirror circuits.
- Verified VREF stability across PVT corners and 40°C to 125°C.

Two-Stage Operational Transconductance Amplifier

Analog IC Design Lab | Prof. Ashudeb Dutta | Mar 2026

- Designed a two-stage single-ended OTA for VDD = 1.8 V using GPDK 45 nm CMOS in Cadence Virtuoso.
- Performed DC, AC, transient, slew-rate, and noise analyses.
- Achieved 65 dB DC gain, 87.9 MHz UGB, and 76.92° phase margin.

RELEVANT COURSES

Analog IC Design, Analog IC Design Lab, Semiconductor Devices & Modeling, VLSI Technology, Circuit Theory, Analog Electronics, CMOS Digital VLSI Design, Digital IC Design Lab, Nanoelectronics Devices & Principles, Memory Devices & Technology.

CERTIFICATIONS

NPTEL Elite Certification -- Op-Amp Practical Applications: Design, Simulation and Implementation -- 72%

POSITIONS OF RESPONSIBILITY

Teaching Assistant -- NPTEL: Basic Electrical Circuits

Under Dr. Gajendranath Chaudhury | IIT Hyderabad

LANGUAGES

- English
- Telugu
- Hindi